Identifying Monetary Policy Shocks via Changes in Volatility: A replication with Alternative Data

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Abstract

We replicate the analysis by Lanne and Lütkepohl (2008) on monetary policy shock identification by means of the Changes-in-Volatility identification for the same monthly sample period 1965M1–1996M12. The CV-SVAR approach with two regimes (1979M10 and 1984M2) confirms the presence of statistically significant heteroskedasticity between regimes. Contrary to the original paper, however, both NBR and NBR/TR schemes are rejected by likelihood ratio tests on over-identifying restrictions ($p = 0.0012$ and $p = 0.0017$). This discrepancy arises because no available R package implements the full AB-SVAR likelihood jointly with CV-identification; replicating the exact estimation procedure would require custom implementation of the iterative algorithm detailed in their Appendix. Although theoretical schemes are rejected by the data, the estimated IRFs do not differ significantly between specifications, suggesting that the overall long-run dynamics are not massively influenced by contemporaneous interaction.

1. Introduction

The identification of monetary policy shocks in structural VAR models (SVAR) has traditionally been at the centre of a broad academic debate. The main obstacle resides in the fact that monetary market models do not usually provide enough theoretical restrictions to uniquely identify structural shocks, making a formal comparison between alternative schemes *de facto* impossible (Christiano et al., 1999; Bernanke, 1990).

Lanne and Lütkepohl (2008) (from now onwards LL) propose a possible solution: exploiting changes in the volatilities of the structural shocks as a source of additional identifying information. If the covariance matrix of reduced-form residuals changes not proportionally between regimes, the volatility ratios allow a simultaneous diagonalization of the covariance matrices of the different regimes, obtaining the correct impact matrix with no need of further theoretical assumptions (Rigobon, 2003). The economic restrictions become then *over-identifying* and statistically testable.

The present paper replicates the one by LL using IP and CPI instead of real GDP (interpolated from quarterly data to monthly) and GDP deflator, for the same sample period 1965M1–1996M12.

2. Model and identification

2.1 The AB-SVAR model

The general setup of LL is an AB-SVAR model in the notation of Amisano and Giannini (1997):

$$\mathbf{A}\, y_t = C_1 y_{t-1} + \cdots + C_p y_{t-p} + \mathbf{B}\, \varepsilon_t, \tag{1}$$

where y_t is a K-dimensional vector of observable variables, $\varepsilon_t \sim (0, I_K)$ are the structural innovations, and $\mathbf{A}$, $\mathbf{B}$, C_i are matrices $(K \times K)$. The reduced-form error is $u_t = \mathbf{A}^{-1}\mathbf{B}\, \varepsilon_t$.

Variables are partitioned into non-policy variables $y_{1t} = (\mathrm{ip}_t,\, \pi_t,\, \mathrm{pcom}_t)'$ – industrial production, CPI, commodity prices – and a monetary block $y_{2t} = (\mathrm{TR}_t,\, \mathrm{NBR}_t,\, \mathrm{FF}_t)'$ – total reserves, non-borrowed reserves, federal funds rate. This partition implies

$$\mathbf{A} = \begin{bmatrix} I_3 & 0 \\ -A_{21} & I_3 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} B_{11} & 0 \\ 0 & B_{22} \end{bmatrix}, \tag{2}$$

where the matrix A_{21} is estimated by standard OLS from the reduced equation of the monetary variables:

$$y_{2t} = A_{21}\, y_{1t} + \sum_{i=1}^{p} C_{2,i}\, y_{t-i} + v_{2t}. \tag{3}$$

Identifiyng monetary shocks requires a bijective map from $E(v_{2t}v_{2t}') = B_{22}B_{22}'$ to B_{22}.

2.2 Identification via Changes in Volatility

In agreement with LL, it is assumed that the covariance mutates in correspondence of two structural breaks T_{B1} and T_{B2}:

$$E(v_{2t}v_{2t}') = \begin{cases} \Sigma_1 = B_{22}B_{22}' & t < T_{B1} \\ \Sigma_2 = B_{22}\Omega_1 B_{22}' & T_{B1} \le t < T_{B2} \\ \Sigma_3 = B_{22}\Omega_2 B_{22}' & t \ge T_{B2} \end{cases} \tag{4}$$

where $\Omega_j = \mathrm{diag}(\omega_{j1}, \omega_{j2}, \omega_{j3})$ with positive elements which measure the relative volatility ratios across regimes.

2.3 Identification schemes

Following Bernanke (1990) and Christiano et al. (1999), LL derive an explicit parameterization of B_{22} in terms of money market structural parameters $\psi = (\alpha, \beta, \gamma, \phi^d, \phi^b, \sigma_d, \sigma_b, \sigma_s)'$. The four principal schemes are:

- **FF**: $\phi^d = 1/(1-\gamma)$, $\phi^b = -\phi^d$ – Fed induces shocks through the federal funds rate (Bernanke, 1990).
- **NBR**: $\phi^d = \phi^b = 0$ – shocks propagate through non-borrowed reserves; these latter do not instantly respond to neither demand nor policy shocks (Christiano et al., 1999).
- **NBR/TR**: $\alpha = \phi^b = 0$ – Fed entirely accomodates demand shocks to total reserves (Strongin, 1995).
- **BR**: $\phi^d = 1$, $\phi^b = \alpha/\beta$, $\gamma = 0$ – targeting of borrowed reserves (Cosimano and Sheehan, 1994).

$$\mathsf{B}_{22} = \begin{bmatrix} \sigma_d \dfrac{\beta - \phi^d\alpha\gamma + \phi^d\alpha}{\beta+\alpha} & -\alpha\sigma_s \dfrac{\gamma-1}{\beta+\alpha} & -\alpha\sigma_b \dfrac{-1+\phi^b\gamma-\phi^b}{\beta+\alpha} \\ \sigma_d\phi^d & \sigma_s & \sigma_b\phi^b \\ \sigma_d \dfrac{\phi^d\gamma - \phi^d + 1}{\beta+\alpha} & \sigma_s \dfrac{\gamma-1}{\beta+\alpha} & \sigma_b \dfrac{-1+\phi^b\gamma-\phi^b}{\beta+\alpha} \end{bmatrix}$$

Since our goal is to perform LR tests with our dataset, and not to derive the entire parameterization via maximum likelihood estimation as done by LL, we only test two of the four schemes – NBR and NBR/TR – which can better be thought as short-run restrictions on the monetary impact submatrix: in terms of B_{22},the NBR scheme imposes $B_{22}[2,1] = B_{22}[2,3] = 0$ (NBR row orthogonal to demand and policy shocks), whereas NBR/TR requires $B_{22}[1,2] = B_{22}[1,3] = B_{22}[2,3] = 0$ (recursive).

3. Data and estimation

We use a monthly sample 1965M1–1996M12 ($T = 384$) for the United States, coherently with LL. All data, except the commodity prices which comes from the World Bank Pink Sheet, are downloaded by FRED; the non-policy variables are `INDPRO`, `CPIAUCSL` and the commodity price index; the monetary variables are `TOTRESNS`, `BOGNONBR` and `FEDFUNDS`. All variables but the FF rate are in logs.

The model is fitted with $p = 13$ lag, as in LL.[1] CV identification is applied is applied to the monetary sub-VAR(13) through the `svars` package in R. The structural breaks are fixed at 1979M10 (beginning of the Volcker era) and 1984M2 (ending of the Volcker era), in accordance with the original paper. Even though we maintain coherence with LL and estimate a VAR in log-levels, we notice that the variables included all fail the Augmented Dickey Fuller (ADF) test for cointegrated variables, providing strong evidence of a unit-root as also hinted by figure 1. For this reason, in the Appendix we report the plot and the same tables present in the paper for the first-differentiated system, which fixes co-integration and ensures the model's stability. Although all VAR models here considered fail diagnostic tests (Kilian and Lütkepohl, 2017) such as the PT test for residuals' serial correlation, JB test for nonnormality - as expected, since heteroskedasticity in general implies non-Gaussianity (Lewis, 2025) -, we relate this to the presence of strong heteroskedasticity in the time series as confirmed also by an ARCH test. In order not to alter the variance structure essential for a successful heteroskedastic identification, we decide to not include dummy variables for shocks or other data-cleaning methods, except to possibly account for seasonality.

4. Results

4.1 Structural break validation

Table 1 reports LR tests for each break significance, conditional to the presence of the other break. Both breaks

[1]Information criteria AIC and FPE would suggest $p = 3$; we adopt $p = 13$ to guarantee comparability with the original. The same analyses which follow are performed on this short-lag specification, with no relevant changes in the final results.

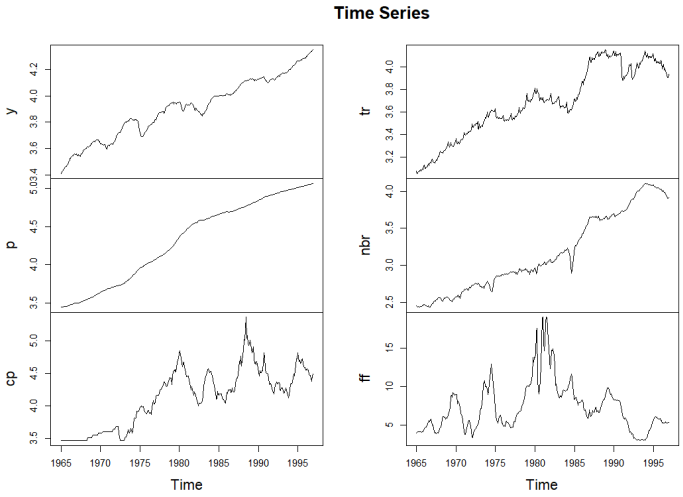


Figure 1: Time series employed in the replication.

appear to be highly significant: the second break of 1984M2 contributes with LR= 229.6 and the first one of 1979M10 with LR= 107.9 (d.f.= 3 , in both cases, $p < 0.001$). The results confirm those of LL and show that the heteroskedasticity structure between the three regimes is significant and robust.

Table 1: LR test for the correct individuation of the structural breaks

Null hypothesis H_0	Stat. LR	p-value
$\Sigma_1{=}\Sigma_2 \mid \Sigma_2{\neq}\Sigma_3$	107.85	<0.001
$\Sigma_2{=}\Sigma_3 \mid \Sigma_1{\neq}\Sigma_2$	229.56	<0.001

Note: d.f.= 3; LR statistic asymptotically distributed as a $\chi^2(3)$.

Table 2 presents estimates of $\hat{\Omega}_1$ (Volcker regime) and $\hat{\Omega}_2$ (post-Volcker) for the tri-regime unrestricted model. The federal funds rate suffers the most pronounced volatility change: $\hat{\omega}_{\text{FF}} = 11.82$ in the Volker regime against $\hat{\omega}_{2,\text{FF}} = 0.285$ for the post-Volcker one, reflecting the radical shift of FED's operating regime. A Wald test for $\omega_1 = \omega_2$ (TR vs. NBR in the Volcker regime) does not reject the null hypothesis of equal volatility ratios ($p = 0.50$), whereas all comparisons with the FF are strongly significant; in the post-Volcker regime, the most relevant one in LL, all pairs are instead statistically distinguishable.

Table 2: Estimates of $\hat{\Omega}_1$ and $\hat{\Omega}_2$, unrestricted model

	$\hat{\Omega}$ (Volcker)		$\hat{\Omega}_2$ (post-Volcker)	
	Estimate	(s.e.)	Estimate	(s.e.)
TR	1.602	(0.458)	4.337	(0.744)
NBR	1.236	(0.267)	1.320	(0.203)
FF	11.815	(2.420)	0.285	(0.045)

Note: Variances in the pre-Volcker regime (1965M1–1979M9) are normalized to 1.

4.2 Over-identifying restrictions test

Table 3 displays LR tests for the NBR and NBR/TR schemes. Both are rejected at the 1% confidence level: NBR with a LR= 13.46 ($p = 0.0012$, d.f.= 2), whereas NBR/TR with a LR= 15.10 ($p = 0.0017$, d.f.= 3).

Table 3: LR test of over-identifying restrictions

Scheme	H_0	LR	d.f.	p-val.
NBR	$\phi^d{=}\phi^b{=}0$	13.463	2	0.0012
NBR/TR	$\alpha{=}\phi^b{=}0$	15.102	3	0.0017

Note: Both rejected at $p<0.01$.

This is the main result that distinguishes the present replication work from the original by LL, where the NBR/TR scheme by Strongin (1995) was not rejected by the data ($p = 0.203$) and was instead chosen as the privileged money market model for the analysed period. The rejection of the NBR/TR scheme may suggest that the assumption that Fed completely accomodates total reserves demand shocks ($\alpha = \phi^b = 0$) is not supported by the data when one uses industrial production instead of real GDP as a measure of economic activity and CPI instead of the deflator as a measure of inflation[2].

Table 4 shows estimates of $\hat{B}_{22}$ for both models. In the NBR scheme, the NBR row displays zeros in column 1 and column 3 by construction; in the NBR/TR scheme the TR row has zero restrictions in column 2 and column 3, and position $(2, 3)$ is also fixed at null value. Unrestricted estimates, however, seem to be coherent between the two hypotheses and with the uncostrained model, which suggests that statistical rejection is not driven by structurally divergent mechanisms, but by a genuine tension with respect to the identified heteroskedastic scaffolding.

Table 4: $\hat{B}_{22}$ matrix for NBR and NBR/TR

	NBR			NBR/TR		
	ε^{tr}	ε^{nbr}	ε^{ff}	ε^{tr}	ε^{nbr}	ε^{ff}
TR	0.0124	0.0018	0.0018	0.0128	0.0000	0.0000
NBR	0.0000	0.0154	0.0000	0.0017	0.0153	0.0000
FF	0.0129	−0.0451	0.4050	0.0181	−0.0497	0.3996

Note: Columns correspond to reserve demand shocks (ε^{tr}), to non-borrowed reserves(ε^{nbr}) and to policy shocks (ε^{ff}). 0.0000 values are necessary placeholders for the imposed restrictions.

4.3 Impulse responses

Figure 2 shows the impulses responses of TR, NBR and FF to a monetary policy shock, for the unrestricted model (with wild-bootstrapped confidence bands at 95%) and for the two schemes NBR and NBR/TR.

[2]The divergence from LL in LR test outcomes reflects the use of a B model in the CV decomposition. Following an AB approach would require implementing LL's complex iterative GLS estimator, as unfortunately no R function applies CV id. to an AB setting.

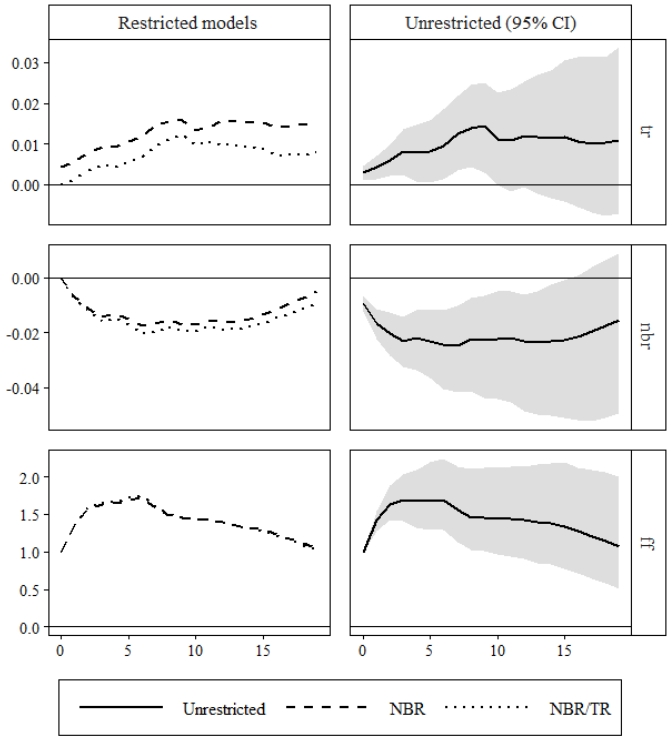


Figure 2: Impulse responses to a monetary policy shock. *Left*: NBR (dashed) and NBR/TR (dotted). *Right*: bootstrapped unconstrained model with confidence bands(grey area).

Notwithstanding the formal rejection of both identification schemes, the corresponding impulse responses remain within the unrestricted model's confidence bands for the entirety of the considered horizon (20 months). Notably, non-borrowed reserves display a persistent and significative negative response (approx. −0.02 in logarithmic terms after 5 months), coherently with a restrictive shock to the Federal Funds rate as shown in the hunched function of the bottom panel. These patterns, though economically coherent, are not sufficient to allow the schemes to overcome formal over-identifying tests. All estimated IRFs suffer from high persistence due to cointegrated variables. See the Appendix for the ones obtained after taking first differences of the whole system.

5. Conclusions

We replicated LL2008, simplifying its most complex mechanisms at the cost of some coherence of the results, for the same sample 1965M1–1996M12. Here are the most interesting findings.

First, the presence of two structural breaks in the volatilities of the monetary shocks (1979M10 e 1984M2) is strongly confirmed, even with two different variables ($p < 0.001$), with the 1984M2 break clearly more significant and pronounced. The Federal Funds rate suffers the biggest change in volatility, with $\hat{\lambda}_{\text{FF}}$ dropping from 11.82 in the Volcker regime to 0.285 in the post-Volcker.

Second, both the NBR and the NBR/TR schemes are rejected ($p = 0.0012$ e $p = 0.0017$), differently from LL2008

where NBR/TR was not rejected by the data.

Third, the impulse responses of the two schemes remain within the confidence bands of the benchmark model, which implies that the formal rejection of the two schemes does not translate into radically different economic implications in the long-run.

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APPENDIX

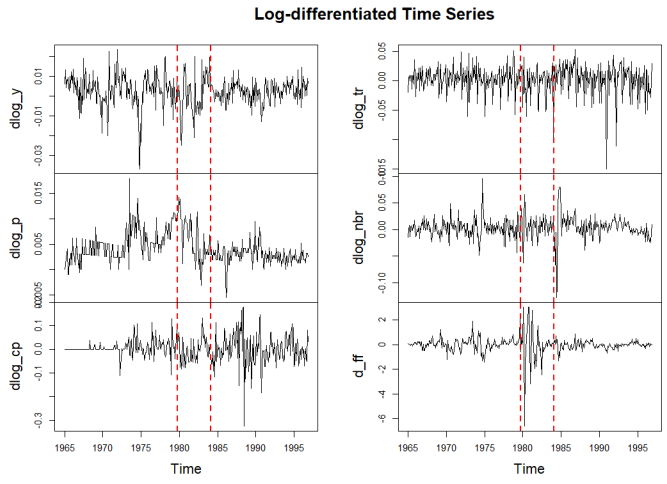


Figure 3: ADFs for monetary ts: H_0: unit root; p-vals < 0.01.

The following tables replicate Tables 1–4 of the main text for the first-differenced system, which removes the unit-root component and guarantees stability of all estimated VAR models. ADF tests on the transformed monetary series all reject the null of a unit root at $p < 0.01$ (Figure 3). The overall conclusions are qualitatively unchanged.

Table 5: LR test for structural breaks – differenced system

Null hypothesis H_0	Stat. LR	p-value
$\Sigma_1{=}\Sigma_2 \mid \Sigma_2{\neq}\Sigma_3$	250.33	<0.001
$\Sigma_2{=}\Sigma_3 \mid \Sigma_1{\neq}\Sigma_2$	124.54	<0.001

Table 6: Estimates of $\hat{\Omega}_1$ and $\hat{\Omega}_2$, unrestricted model – differenced system

	$\hat{\Omega}$ (Volcker)		$\hat{\Omega}_2$ (post-Volcker)	
	Estimate	(s.e.)	Estimate	(s.e.)
TR	1.292	(0.347)	3.560	(0.553)
NBR	1.507	(0.397)	1.374	(0.212)
FF	14.961	(3.704)	0.298	(0.047)

Table 7: LR test of over-identifying restrictions – differenced system

Scheme	H_0	LR	d.f.	p-val.
NBR	$\phi^d{=}\phi^b{=}0$	16.568	2	< 0.001
NBR/TR	$\alpha{=}\phi^b{=}0$	19.202	3	< 0.001

Table 8: $\hat{B}_{22}$ matrix for NBR and NBR/TR – differenced system

	NBR			NBR/TR		
	ε^{tr}	ε^{nbr}	ε^{ff}	ε^{tr}	ε^{nbr}	ε^{ff}
TR	0.0123	0.0018	0.0018	0.0127	0.0000	0.0000
NBR	0.0000	0.0156	0.0000	0.0017	0.0156	0.0000
FF	0.0107	−0.0458	0.4108	0.0165	−0.0506	0.4052

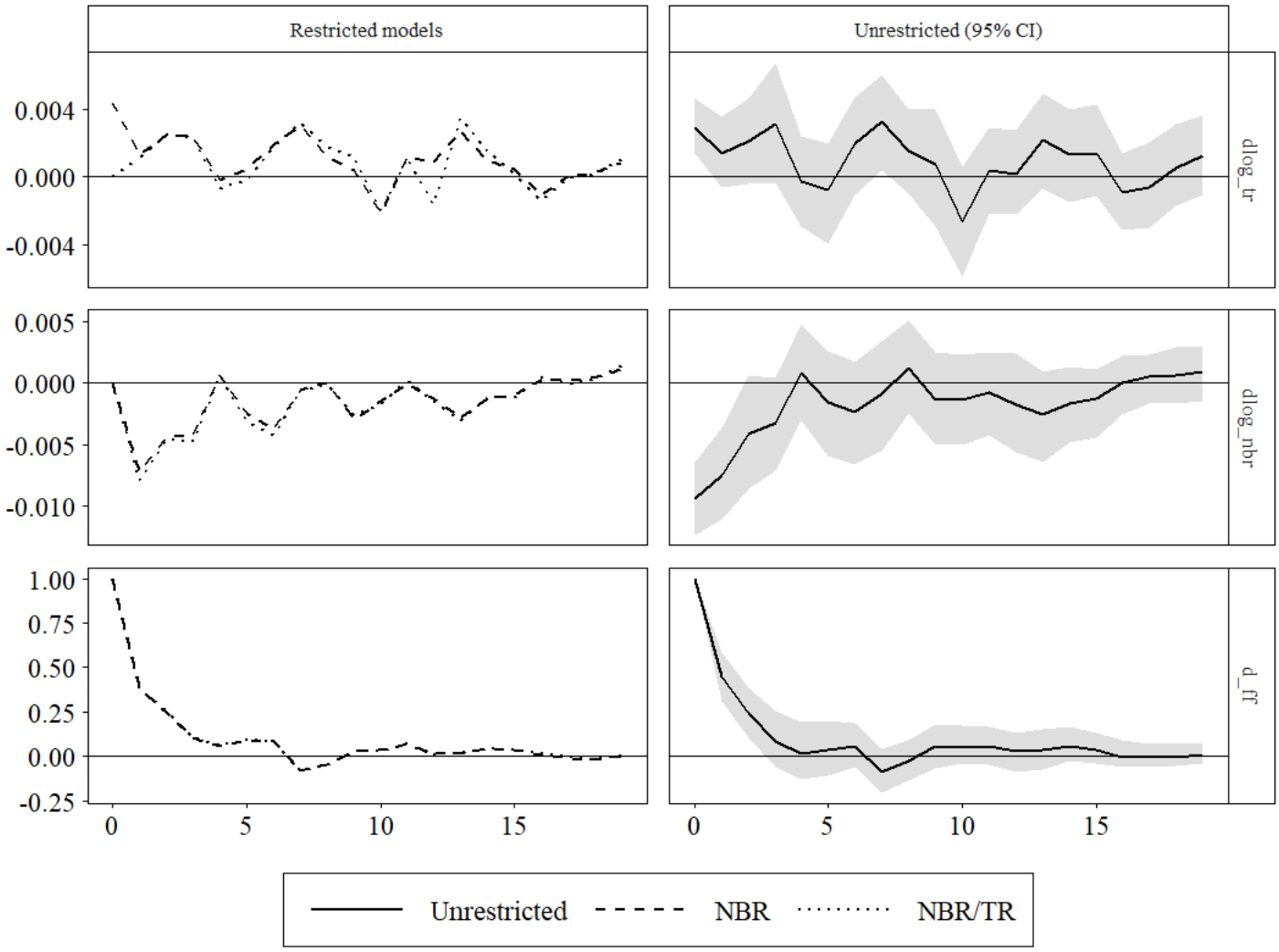


Figure 4: Impulse responses to a monetary policy shock – differenced system ($p = 13$). *Left*: NBR (dashed) and NBR/TR (dotted). *Right*: bootstrapped unrestricted model with 95% confidence bands. The stationary specification removes the persistence artifact visible in Figure 2